# Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

| Role | Name | Email |
|---|---|---|
| **Developer (You)** | Lincoln Crowley | crowleyl@udel.edu |
| **Friend (optional)** | | |

## Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

| |
|---|
| I want to make a monthly budget and expense tracker for college students. It is really easy to lose track of how much money you spend on food, school supplies, and going out with friends. Knowing this, my goal is to build a simple site where students can log what they buy and see if they are staying under their monthly budget. |

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

| |
|---|
| Some existing apps and websites that do this are Rocket Money, YNAB (You Need A Budget), and Goodbudget. |

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

| |
|---|
| My Drafter website will have 5 pages and at least 7 routes where a student can set their monthly spending limit, add new expenses with a price and category, filter their purchases, and see how much money they have left for the month. |

# Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The program takes the user's monthly budget and expense info from input boxes, saves each item in a list of Expense dataclass instances inside State, uses a for-loop and if-statements to add up total spending and check the budget limit, and shows a summary page with how much money is left.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will type their monthly budget amount and each expense's name and price into TextBox inputs, pick a spending category (Food, School, Rent, or Fun) from a SelectBox, and click a CheckBox if the purchase was an essential need.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program uses a for-loop to go through the list of Expense objects in State and add up all the costs. It also filters the list by category or essential status, and uses if statements to check if the total spent is under budget, close to the limit, or over budget.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user gets back a summary page showing their total money spent, how much budget they have left, a filtered list of their logged expenses, and a warning message if they are close to or over their monthly budget.

# Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

The main things (nouns) my program needs to track are the overall "Budget" (in the main State dataclass) and each "Expense" (stored as a list of Expense dataclass instances inside State).

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need for Expense: name (str), category (str), cost (float), and is_essential (bool).
Details I will ignore: the time of day I bought it, store street address, cashier name, credit card number, and item barcode.

# Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The hardest parts will be making sure the State updates properly across all 7 routes and 5 pages, converting string inputs from a TextBox into floats without crashing if someone types letters, and writing unit tests for all my routes.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

If I use an LLM for help, I will write my own dataclass fields and function headers first. I will not copy and paste code I do not understand, and I will write my own helper functions and assert_equal unit tests to make sure everything works the way we learned in class.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My main goal is to build a working Drafter website that hits all the CISC106 requirements and is actually useful for tracking my own spending this semester.

# Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool in Python using Drafter to help college students track their monthly budget and expenses. What are 3 common problems students have with budgeting, and what are 3 simple features my tool should have?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The AI said students usually forget small daily purchases like coffee, mix up essential needs with wants, and do not realize they are over budget until the end of the month. It mentioned Rocket Money and YNAB, and suggested adding an expense logger, a category breakdown, and automatic bank syncing with AI receipt scanning.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

| |
|---|
| Automatic bank account syncing and AI receipt scanning are way too complicated for an intro CS1 project in Drafter. We only use basic inputs like Python dataclasses. What is a much simpler feature I can do instead? |

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

| |
|---|
| Talking it through helped me keep my idea simple and realistic. Instead of doing anything complicated with bank accounts or photos, I will just have people type expenses into a TextBox, pick a category from a SelectBox, and click a CheckBox if it is essential, then loop through the list of Expense objects. |

# Part 6: Resource Planning

1. What resources will you use if you get stuck?

| |
|---|
| I will check the Drafter documentation, look back at our Comp Sci BlockPy assignments and lecture notes, go to TA office hours, and use the class Drafter planning LLM. |

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

| |
|---|
| 1. Read the traceback error message and line number from the bottom up to see what broke. 2. Print out my State fields or test my helper functions by themselves using assert_equal. 3. Check that every @route function takes State as its first parameter, returns a Page object, and has parameter names that match my input boxes. 4. If it fails, check main.py for syntax errors or broken imports. |

# Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

With Gemini:

Me: I want to build a simple web tool in Python using Drafter to help college students track their monthly budget and expenses. What are 3 common problems students have with budgeting, and what are 3 simple features my tool should have?

AI: Here are 3 common budgeting problems college students face:
1. Small Purchases Adding Up: Students often forget to track small daily buys like coffee, snacks, or dining hall extras.
2. Needs vs. Wants Confusion: It is hard to tell how much money is going toward essential school/living costs versus fun spending.
3. Late Realization: Students often only check their balance after they have already overspent for the month.
Existing apps that do this include Rocket Money, YNAB (You Need A Budget), and Goodbudget.
3 features your tool could have:
1. Quick Expense Logger: Let users enter the item name, cost, and category.
2. Budget Status Warning: Show how much money is left and warn them if they pass 80% or 100% of their budget.
3. Automatic Bank Sync & AI Receipt Scanner: Connect to their bank API and scan photos of receipts automatically.

Me: Automatic bank account syncing and AI receipt scanning are way too complicated for an intro CS1 project in Drafter. We only use basic inputs like TextBox, SelectBox, and CheckBox with Python dataclasses. What is a much simpler feature I can do instead?

AI: Good point! For a CS1 Drafter project, instead of bank syncing or receipt scanning, you can add a simple CheckBox when logging an expense to mark whether it is "Essential" (a need vs. a want) and a SelectBox to filter expenses by category (like Food, School, Rent, or Fun). You can store each item in an Expense dataclass inside a list in State, and use a for loop to add up how much was spent on needs versus wants.

# Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again.

Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.